

IIT JAM Mathematics

Group Theory Cheat Sheet

1. Groups: Basics

Group $(G, *)$: A set G with a binary operation $*$ satisfying:

1. **Closure**: $a * b \in G \quad \forall a, b \in G$.
2. **Associativity**: $a * (b * c) = (a * b) * c$.
3. **Identity**: $\exists e \in G$ s.t. $a * e = e * a = a$.
4. **Inverse**: $\forall a \in G, \exists a^{-1}$ s.t. $a * a^{-1} = e$.

Abelian Group: $a * b = b * a \quad \forall a, b \in G$.

Non-Abelian Group: $\exists a, b \in G$ s.t. $a * b \neq b * a$.

Order of G ($|G|$): Number of elements in G .

Order of $a \in G$ ($|a|$): Smallest $n \in \mathbb{N}$ s.t. $a^n = e$.

2. Cyclic Groups

Definition: $G = \langle a \rangle = \{a^n \mid n \in \mathbb{Z}\}$ for some generator $a \in G$.

Properties:

- Every cyclic group is Abelian.
 - Every subgroup of a cyclic group is cyclic.
 - If $|G| = n$ and $G = \langle a \rangle$, then a^k generates G iff $\gcd(k, n) = 1$.
 - Number of generators of a finite cyclic group of order n is $\phi(n)$ (Euler's totient function).
 - The order of an element a^k in $\langle a \rangle$ of order n is $|a^k| = \frac{n}{\gcd(n, k)}$.
 - For every divisor d of n , there is exactly one subgroup of order d . The number of elements of order d is $\phi(d)$.
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3. Permutation Groups

Symmetric Group (S_n) : The group of all bijections (permutations) on a set of n elements. $|S_n| = n!$.

Properties:

- S_n is non-abelian for $n \geq 3$.
- **Cycle**: A permutation that shifts a subset of elements in a cyclic fashion.
- **Disjoint cycles** commute: If α, β are disjoint, $\alpha\beta = \beta\alpha$.

- **Order of a permutation**: LCM of the lengths of its disjoint cycles.

Transpositions & Alternating Groups:

- **Transposition**: A cycle of length 2.
 - Every permutation can be written as a product of transpositions.
 - **Even/Odd**: A permutation is even (odd) if it is a product of an even (odd) number of transpositions.
 - **Alternating Group** (A_n) : Set of all even permutations in S_n . Subgroup of S_n with order $|A_n| = \frac{n!}{2}$.
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4. Lagrange's Theorem & Subgroups

Subgroup $(H \leq G)$: A subset H of G is a subgroup if it is a group under G 's operation.

- **One-step Test**: $H \neq \emptyset, ab^{-1} \in H \quad \forall a, b \in H$.

Cosets: Left coset $aH = \{ah \mid h \in H\}$, Right coset $Ha = \{ha \mid h \in H\}$.

Index $([G : H])$: Number of distinct left (or right) cosets of H in G .

Lagrange's Theorem

If G is a finite group and $H \leq G$, then the order of H divides the order of G .

$$|G| = |H| \cdot [G : H]$$

Corollaries:

- The order of any element $a \in G$ divides $|G|$.
 - $a^{|G|} = e$ for all $a \in G$.
 - Every group of prime order is cyclic.
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5. Normal Subgroups & Quotient Groups

Normal Subgroup $(N \triangleleft G)$: A subgroup N is normal if its left and right cosets coincide:

$$gN = Ng \quad \text{or} \quad gNg^{-1} = N \quad \forall g \in G$$

Properties of Normal Subgroups:

- Every subgroup of an Abelian group is normal.
- Any subgroup of index 2 is normal.
- The center of a group $Z(G) = \{x \in G \mid xg = gx \quad \forall g \in G\}$ is always a normal subgroup.

Quotient Groups (Factor Groups)

If $N \triangleleft G$, the set of left cosets $G/N = \{gN \mid g \in G\}$ forms a group under $(aN)(bN) = (ab)N$.

- **Order**: $|G/N| = \frac{|G|}{|N|}$.
 - If G is abelian, G/N is abelian.
 - If G is cyclic, G/N is cyclic.
 - If $G/Z(G)$ is cyclic, then G is abelian.
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6. Group Homomorphisms

Homomorphism: A map $\phi : G \rightarrow G'$ preserving the group operation:

$$\phi(ab) = \phi(a)\phi(b) \quad \forall a, b \in G$$

Properties:

- $\phi(e) = e'$ (maps identity to identity).
- $\phi(a^n) = (\phi(a))^n$.
- If $|a|$ is finite, $|\phi(a)|$ divides $|a|$.

Kernel and Image

Kernel: $\ker(\phi) = \{g \in G \mid \phi(g) = e'\}$.

Note: $\ker(\phi)$ is a normal subgroup of G .

Image: $\text{Im}(\phi) = \{\phi(g) \mid g \in G\}$.

Note: $\text{Im}(\phi)$ is a subgroup of G' .

Injectivity: ϕ is 1-1 (injective) $\iff \ker(\phi) = \{e\}$.

Isomorphism Theorems

First Isomorphism Theorem: If $\phi : G \rightarrow G'$ is a homomorphism, then:

$$G / \ker(\phi) \cong \text{Im}(\phi)$$

Specifically, if ϕ is onto (surjective), then $G / \ker(\phi) \cong G'$.